Ghiyath Abd Alrazzaq

CAD/CAM Engineer | CNC Programmer | Mechanical Design Engineer

Damascus , Syria | ghiyath.gg.1@gmail.com | +963930348565 | Portfolio

SUMMARY

Detail-oriented Mechanical Engineering graduate with hands-on experience in CAD/CAM engineering CNC programming and mechanical design Skilled in 3-axis toolpath generation using **Autodesk PowerMill**, roughing rest machining finishing strategies and G-code simulation. Proven track record in designing assembling and testing a physical 3-axis CNC machine integrating hands-on manufacturing knowledge with **SolidWorks** 3D modeling and GD&T standards Passionate about translating complex engineering designs into optimized efficient and precise CNC machining workflows.

PROJECTS

3-Axis CNC Machine Design, Fabrication & Programming | *Graduation Project, Damascus University*

- Designed the complete mechanical structure of a 3-axis CNC router in SolidWorks, validating structural rigidity and dimensional accuracy.
- Selected and calculated drive components, linear guides, lead screws, couplings, and stepper motors based on load and precision requirements.
- Physically built aligned and calibrated the complete CNC machine verifying real-world tolerances, motion clearance, and assembly fits against CAD models.
- Configured and integrated a GRBL-based controller to execute G-code driven 3-axis motion profiles.
- Applied practical CNC manufacturing concepts and toolpath verification to validate the physical machinability and operational reliability of all custom parts.

Plastic Injection Mold Design | *4th-Year Academic Project, Damascus University*

- Designed a multi-part plastic injection mold including core cavity side cores and parting lines using SolidWorks.
- Analyzed PMMA polymer properties to establish design allowances for material shrinkage, thermal behavior, and dimensional tolerances.
- Incorporated critical design for manufacturability (DFM) features such as draft angles, uniform wall thickness and automated part ejection mechanisms.

SKILLS

CAM & CNC Programming:

- Autodesk PowerMill
- 3-Axis CNC Programming
- Toolpath Generation & Strategy Selection (Roughing, Rest Machining, Finishing)
- Toolpath & Machining Simulation
- G-code Generation & Verification
- Fanuc CNC Controller Fundamentals

CAD & Engineering Graphics:

- SolidWorks (3D Modeling, Assembly, Sheet Metal, Shop Drawings)
- AutoCAD
- Geometric Dimensioning & Tolerancing (GD&T Basics)

Mechanical & Manufacturing Engineering:

- CNC Machining Processes & Setup
- Design for Manufacturing & Assembly (DFM / DFA)
- Mechanical Assembly & Component Selection
- Plastic Injection Mold Design Basics

EDUCATION

- **BSc in Mechanical Engineering – Design & Production Engineering** | Damascus University | 2026

LANGUAGE

- **Arabic :** Native Speaker
- **English :** Professional Working Proficiency

PORTFOLIO

- **Engineering & Design Portfolio:** https://eng-ghiyath.github.io